

capital (K) from adjusted value-added (J) gives us the surplus-value itself (L).

$$(J)-(K)=(L)$$

Footnotes

1. This approach generally equalizes the profit-wage ratio (π/w) with the rate of surplus-value -properly speaking the money form of the rate of surplus-value, (s/v). See, for example, (Glynn and Sutcliffe, 1972), (Hodgson, 1974), (Boddy and Crotty, 1975) and (Rowthorn, 1976).
2. These are positions that I do not agree with; however I do not intend to discuss them here. For a general theoretical and empirical critique of the first approach, see (Shaikh, 1978A).
3. All three studies are related to the U.S.A. economy. The former two measure the rate of profit while the third study only measures the rate of surplus-value. In this context, Shaikh states that (Yaffe, 1976 and (MFZ, 1977) note only the existence of the problem. See (Shaikh, 1978A, p.2). It is worthwhile to mention that there exists a study on the rate of surplus-value in Russia in 1908 done by Lenin in 1912. His measurement for the manufacturing sector was around 100 per cent. See (Manuel d'Economie, Edition Sociales, Paris:1956, p.126; quoted in Somer, 1965, p.6).
4. I will not discuss the fixed and circulating parts of means of production since they require discussion of some other theoretical questions such as, turnover-time, etc.
5. See (Marx, 1967, p.167; quoted by Shaikh, 1977, p.119).
6. "Although the rate of surplus-value is an exact expression for the degree of exploitation of labor-power, it is in no sense, an expression for the absolute amount of exploitation. For example, if the necessary labor -5 hours and the surplus-labor -5 hours, the degree of exploitation is 100 per cent. The amount of exploitation is here measured by 5 hours. If, on the other hand, the necessary labor -6 hours and the surplus-labor -6 hours, the degree of exploitation remains, as before, 100 per cent, while the actual amount of exploitation has increased 20 per cent, namely from five hours to six." (Marx, 1954, pp. 209-210).

Table I.

CALCULATION OF THE VARIABLE CAPITAL--1971A. Manufacturing

1. Productive Wage Workers--PWMAN 1	13.544
(ETRP, Table C-2, p.212)	
2. Unproductive Wage Workers--UPWMAN 1	5.079
(Ibid.)	
3. (1) / (1+2).....	.73
4. Self-Employed Persons--(SEMPMAN).....	0.248
(NIPA, TABLE 6.10B, p.264, 1.7)	
5. Productive Self-Employed Persons--(PSEMPMA)....	0.18
(3) x (4)	
6. Total Number of Productive Workers--(PWMAN)....	13.724
(1) + (5)	
7. Total Number of Unproductive Workers--(UPWMAN)..	5.147
(2) + (4-5)	
8. Productive Workers in Labor Force.....	.73
(6) / (6+7)	
9. Employer Contribution for Social Ins.--(ECSECMAN)	8.098
(NIPA, Table 6.14, p.272, 1.7)	
10. Productive Workers' Share in ECSECMAN.....	5.91
(8) x (9)	
11. Average Weekly Earnings (in \$)--(AWEMAN).....	142.4
(ETRP, Table C-5, p. 215)	
12. Total Wages of Productive Workers--(VMAN).....	107.512
(6) x (11) x (52) + (10)	=====

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1. In this and the following sectoral tables regarding the calculation of the variable capital, number of workers and self-employed persons, employer contribution for social insurance, total wages of productive workers and variable capital are respectively given in millions and billions of dollars.

B. Mining

1. Productive Wage Workers--(PWMIN 1).....	0.455
(ETRP, Table C-2, p.212)	
2. Unproductive Wage Workers--(UPWMIN 1).....	0.154
(Ibid.)	
3. (1) / (1+2).....	.75
4. Self-Employed Persons--(SEMPMIN).....	0.015
(NIPA, Table 6.10B, p.264, 1.5)	
5. Productive Self-Employed Persons--(PSEMPMIN)...	0.01
(3) x (4)	
6. Total Number of Productive Workers--(PWMIN)....	0.466
(1) + (5)	
7. Total Number of Unproductive Workers--(UPWMIN).	0.158
(2) + (4-5)	
8. Productive Workers in Labor Force.....	.75
(6) / (6+7)	
9. Employer Contrib. for Social Ins.--(ECSECMIN)..	0.287
(NIPA, Table 6.14, p.272, 1.5)	
10. Productive Workers' Share in ECSECMIN.....	0.21
(8) x (9)	
11. Average Weekly Earnings (in \$)--(AWEMIN).....	172.1
(ETRP, Table C-5, p. 215)	
12. Total Wages of Productive Workers--(VMIN).....	4.387
(6) x (11) x (52) + (10)	=====

C. Construction

1. Productive Wage Workers--(PWCON 1).....	3.071
(ETRP, Table C-2, p. 212)	
2. Unproductive Wage Workers--(UPWCON 1).....	0.633
(Ibid.)	
3. (1) / (1+2).....	0.83
4. Self-Employed Persons--(SEMPCON).....	0.719
(NIPA, Table 6.10B, p. 264, 1. 6)	
5. Productive Self-Employed Persons--(PSEMPCON)...	0.60
(3) x (4)	
6. Total Number of Unproductive Workers--(PWCON)..	3.667
(1) + (5)	
7. Total Number of Unproductive Workers--(UPWCON).	0.756
(2) + (4-5)	
8. Productive Workers in Labor Force.....	0.83
(6) / (6+7)	
9. Employer Contribution for Social Ins.--(ECSECCON).	1.958
(NIPA, Table 6.14, p. 272, 1. 6)	
10. Productive Workers' Share in ECSECCON.....	1.623
(8) x (9)	
11. Average Weekly Earnings (in \$)--(AWECON).....	211.7
(ETRP, Table C-5, p. 215)	
12. Total Wages of Productive Workers--(VCON).....	41.993
(6) x (11) x (52) + (10)	=====

D. Transportation, Utilities and Communication

1. Productive Wage Workers--(PWTUC 1)	3.872
(ETRP, Table C-2, p. 212)	
2. Unproductive Wage Workers--(UPWTUC 1)	0.604
(Ibid.)	
3. (1) / (1+2).....	0.87
4. Self-Employed Persons--(SEMPTUC)	0.201
(NIPA, Table 6.10B, p. 264, 1. 10)	
5. Productive Self-Employed Persons--(PSEMPTUC)...	0.17
(3) x (4)	
6. Total Number of Unproductive Workers--(PWTUC)..	4.042
(1) + (5)	
7. Total Number of Unproductive Workers--(UPWTUC).	0.635
(2) + (4-5)	
8. Productive Workers in Labor Force.....	0.86
(6) / (6+7)	
9. Employer Contribution for Social Ins.--(ECSECTUC).2.324	
(NIPA, Table 6.14, p. 272, 1. 10)	
10. Productive Workers' Share in ECSECTUC.....	1.999
(8) x (9)	
11. Average Weekly Earnings (in \$)--(AWETUC)	168.8
(ETRP, Table C-5, p. 215)	
12. Total Wages of Productive Workers--(VTUC).....	37.523
(6) x (11) x (52) + (10)	=====

E. Services

1. Productive Wage Workers--(PWSER or PRPENGPS).... 3.678
(NIPA, Table 6.11B, p. 267, lines 61-62-64-65-66-67)
2. Persons Engaged in Production--(PENGPSER)..... 13.438
(NIPA, Table 6.11B, p. 267, 1. 60)
3. (1) / (2)..... 0.21
4. Employer Contribution for Social Ins.--(ECSECSER). 3.348
(NIPA, Table 6.14, p. 272, 1. 17)
5. Productive Workers' Share in ECSECSER..... 0.70
(3) x (4)
6. Average Weekly Earnings (in \$)--(AWESER) 131.2
(NIPA, Table 6.9B, p. 260, 1. 60)
7. Total Wages of Productive Workers--(VSER)..... 26.009
(1) x (6) x (52) + (5) =====

F. Government Enterprises

1.	Total Number of Non-Governmental Productive Workers--(NONGP).....	26.827
	(PWMAN + PWMIN + PWCON + PWTUC + PWAG)	
2.	Total Numbers of Non-Governmental Unproductive Workers--(NONGUP).....	18.239
	(UPWMAN + UPWMIN + UPWCON + UPWAG)	
3.	(1) / (1+2) -- (PRSNG).....	0.66
4.	Total Numbers of Workers in Government Enterprises-Federal--(NUMWGEF).	0.874
	(<u>NIPA</u> , Table 6.8B, p. 255, 1. 81)	
5.	Total Number of Workers in Government Enterprises-State/Local--(NUMWGESL)	0.521
	(<u>Ibid.</u> , 1.86)	
6.	Productive Workers in Government Enterprises--(PWG).....	0.830
	(3) x (4+5)	
7.	Employer Contribution for Social Insurance--(ECSECGE).....	11.499
	(<u>NIPA</u> , Table 6.14, p. 272, 1.18)	
8.	Total Number of Workers in Government and Government Enterprises (NUMWG).....	10.122
	(<u>NIPA</u> , Table 6.8B, p. 255, 1.76)	
9.	(6) / (8).....	0.08
10.	Productive Workers' Share in ECSECGE.....	0.919
	(9) X (7)	
11.	Average Weekly Earnings (in \$) -- (AWEGE).....	167.0
	(<u>NIPA</u> , Table 6.9B, p. 260, 1.76)	
12.	Total Wages of Productive Workers--(VGE)..	7.866
	(6) x (10) x (52) + (9)	=====

G. Agriculture

1. Productive Workers in Agriculture --(PWAG)..... 1.245
(NIPA, Table 6.8B, p. 255, 1. 4)
2. Unproductive Workers in Agriculture --(UPWAG)... 1.788
3. (1) / (1+2)..... 0.41
4. Employer Contribution for Social Ins.--(ECSECAG). 0.256
(NIPA, Table 6.14, p. 272, 1. 4)
5. Productive Workers' Share in ECSECAG..... 0.11
(3) x (4)
6. Average Weekly Earnings (in \$)--(AWEAG)..... 84.4
(NIPA, Table 6.9B, p. 260, 1. 4)
7. Total Wages of Productive Workers--(VAG)..... 5.567
(1) x (6) x (52) + (5)

H. Variable Capital

1. Total Variable Capital--(VCAP).....230.856
(VMAN + VMIN + VCON + VTUC + VSER + VGE + VAG) =====

Table II.

Computer Names of the Wage Data

PWMAN1: Productive wage workers - Manufacturing
 PWMIN1: Productive wage workers - Mining
 PWCON1: Productive wage workers - Construction
 PWTUC1: Productive wage workers - Transp., Utilities
 and Communic.
 UPWMAN1: Unproductive wage workers - Manufacturing
 UPWMIN1: Unproductive wage workers - Mining
 UPWCON1: Unproductive wage workers - Construction
 UPWTUC1: Unproductive wage workers - Transp., Utilities
 and Communic.
 ECSECMAN: Employer Contribution for Social Insurance -
 Manufacturing
 ECSECMIN: Employer Contribution for Social Insurance -
 Mining
 ECSECCON: Employer Contribution for Social Insurance -
 Construction
 ECSECTUC: Employer Contribution for Social Insurance -
 Tr., Util. & Comm.
 ECSECSER: Employer Contribution for Social Insurance -
 Service
 ECSECAG: Employer Contribution for Social Insurance -
 Agriculture
 ECSECG: Employer Contribution for Social Insurance -
 Govnt. Ent.
 AWEMAN: Average Weekly Earnings - Manufacturing
 AWEMIN: Average Weekly Earnings - Mining
 AWECON: Average Weekly Earnings - Construction
 AWETUC: Average Weekly Earnings - Trans., Util. &
 Comm.
 AWESER: Average Weekly Earnings - Service
 AWEGE: Average Weekly Earnings - Govnt. Ent.
 AWEAG: Average Weekly Earnings - Agriculture
 NUMWGEF: Total Number of Workers in Govnt. Ent. - Federal
 NUMWGESL: Total Number of Workers in Govnt. Ent. - State
 and Local
 NUMWG: Total Number of Workers in Govnt. and Govnt. Ent.
 PRPENGPS: Productive Wage Workers - Service
 PENGPSER: Persons Engaged in Production - Service
 SEMPMAN: Self-employed Persons - Manufacturing
 SEMPMIN: Self-employed Persons - Mining
 SEMPCON: Self-employed Persons - Construction
 SEMPTUC: Self-employed Persons - Trans., Util. & Comm.
 SEMPAG: Self-employed Persons - Agriculture
 PSEPMAN: Productive Self-employed Persons - Manufacture
 PSEPMIN: Productive Self-employed Persons - Mining
 PSEMPCON: Productive Self-employed Persons - Construction
 PSEMPTUC: Productive Self-employed Persons - Trans.,
 Util. & Comm.
 PWAG: Productive Wage Workers - Agriculture